



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Group and Ring Theory
MATP33
Monday, 15 August 2022
Duration: 08:00–13:00

Answers

3. We have U is Artinian but not Noetherian.

4. As

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \{\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \mid \varphi(1) \equiv_n 0\},$$

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/30\mathbb{Z}, \mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}.$$

Here is the full justification: [Either write $\mathbb{Z}_{30} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ and $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \oplus \mathbb{Z}_4$, and work with the easier pieces, or proceed as follows.]

We claim that $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/(m, n)\mathbb{Z}$ as rings. Note that we have an isomorphism

$$\begin{aligned} \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) &\rightarrow \mathbb{Z}/m\mathbb{Z} \\ \varphi &\mapsto \varphi(1). \end{aligned}$$

Moreover, if $\bar{\varphi} \in \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$, it follows that the order of $\bar{\varphi}(1)$ in $\mathbb{Z}/m\mathbb{Z}$ divides n since

$$n\bar{\varphi}(1) = \bar{\varphi}(n) = \bar{\varphi}(0) = 0.$$

The elements of $\mathbb{Z}/m\mathbb{Z}$ whose orders divide n form a subgroup A generated by

$$a = m/\gcd(m, n).$$

(If the order of $x \in \mathbb{Z}/m\mathbb{Z}$ divides n , equivalently $nx \equiv 0 \pmod{m}$, i.e. nx is divisible by m , then $nx/\gcd(m, n)$ is divisible by a , and since $n/\gcd(m, n)$ and a are coprime, the element x is divisible by a .)

Since

$$n\mathbb{Z} \subseteq \mathbb{Z} \twoheadrightarrow \mathbb{Z}/n\mathbb{Z},$$

it follows that

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \{\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \mid \varphi(1) \equiv_n 0\} \cong A.$$

Hence the claim, and thus $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/30\mathbb{Z}, \mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}$. Specifically a generator of $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/30\mathbb{Z}, \mathbb{Z}/12\mathbb{Z})$ is given by the map $1 \mapsto 2$; note that the generator a above is 2 here.

5. The invariant factors are $1, 1, (1-x)^2, (1-x)^2$.

6. We have $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$.

On the other hand, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$.